

Pengfei Hu

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EDUCATION

Stevens Institute of Technology

Sep. 2023 – Present

Ph.D. in Computer Science

Hoboken, NJ

- Advisor: [Dr. Yue Ning](#); GPA: 3.95 / 4.00;
- **Research Interest:** Transfer Learning, Post-Training, Large Language Model, Graph Learning, Interpretability.

University of Southern California

Sep. 2021 – May 2023

M.S. in Analytics

Los Angeles, CA

- GPA: 4.00 / 4.00;
- Research Assistant advised by [Dr. Sze-chuan Suen](#) working on monitoring of diabetes via wearable devices.

Nanjing University of Finance and Economics

Sep. 2016 – Jun. 2020

B.S. in Economic Statistics

Nanjing, China

- GPA: 3.63 / 4.00 (Rank 1st in major, National Scholarship).
- Working on relative deprivation in the digital era ([one publication](#)) supervised by [Dr. Qinghai Li](#).
- Interned at *Alibaba Group* (2021), *Tencent* (2020), *Didi Global* (2019–2020), and *PwC Consulting* (2019).

INTERNSHIP EXPERIENCE

Oak Ridge National Laboratory

May 2025 – Aug. 2025

Research Scientist Intern

Oak Ridge, TN

- **Mentor:** *Ming Fan*; **Leader:** *Dan Lu*;
- **Project 1: Adaptive Graph Transformer for Multi-Reservoir Inflow Forecasting**
 - * Designed an adaptive graph learning module that captures latent inter-reservoir dependencies, coupled with a Transformer encoder to model spatiotemporal inflow dynamics in the Upper Colorado River Basin;
 - * **Paper: First-author** workshop paper at [IEEE International Conference on Data Mining \(ICDM\) 2025](#); extended version (**co-first-author**) published in [Environmental Modelling & Software](#) (IF: 4.6).
- **Project 2: Hydrological Domain-Conditioned Modulation for Cross-Reservoir Generalization**
 - * Conditioned a shared backbone on reservoir meta-attributes (geography, storage, climate) via feature-wise linear modulation for zero-shot transfer to unseen reservoirs;
 - * **Paper: First-author** Oral at [AAAI 2026 Workshop on AI for Environmental Science](#).

SELECTED PROJECTS

Transparent Domain Adaptation via Concept-Grounded Orthogonal Inference

[ICML 2026](#)

- Proposed an orthogonal and concept-grounded inference framework for transparent domain adaptation in clinical event prediction on Electronic Health Records.
- Designed a sparse autoencoder with a dictionary-induced metric to factorize patient representations into label-invariant and domain-covariant components via M -orthogonal residual projection.
- Provided human-understandable explanations via sparse-dimension ablation while achieving superior performance over state-of-the-art DA baselines on two real-world EHR datasets (eICU and OCHIN) under spatial, temporal, and facility-based distribution shifts.

LLM Agent for Causal Discovery in Diagnosis Prediction

[EMNLP 2025 Findings](#)

- Designed an agent-based causal discovery framework leveraging large language models to explore disease relationships and improve interpretability and predictive accuracy in clinical diagnosis tasks.
- Constructed a domain knowledge vector database using ChromaDB, integrating biomedical knowledge from Wikipedia and embedding documents using pre-trained Sentence-BERT models.
- Improved reasoning coverage via RAG over structured domain knowledge.
- Implemented the agentic reasoning pipeline using LangChain and evaluated performance on public EHR datasets.

SKILLS

Foundation Models & Reasoning Techniques: LLM Post-Training (SFT, DPO, GRPO), Test-Time Reasoning, Model Distillation, LLM Agents, Tool-Integrated Reasoning, Retrieval-Augmented Generation (RAG).

Infrastructure & Scalable ML Systems: PyTorch, Hugging Face Transformers, verl, verl-tool, TRL, Accelerate, SGLang, vLLM, PyTorch Geometric, TensorFlow, TensorBoard, Weights & Biases, AWS (EC2, S3).

Agent & Data Infrastructure: LangChain, ChromaDB.

Model Architectures: Transformers, Graph Neural Networks (GNNs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Sparse Auto Encoder.

Programming Languages: Python, R, SQL, Java, Latex.

SELECTED PUBLICATIONS

(The Google Scholar link can be found [here](#), * equal contribution)

First-authored Publications:

1. **Pengfei Hu**, et al., “Exploring Accurate and Transparent Domain Adaptation in Predictive Healthcare via Concept-Grounded Orthogonal Inference”, *International Conference on Machine Learning (ICML 2026)*.
2. **Pengfei Hu**, et al., “Bridging Stepwise Lab-Informed Pretraining and Knowledge-Guided Learning for Diagnostic Reasoning”, *IEEE Journal of Biomedical and Health Informatics (IEEE JBHI, IF: 6.8)*, 2026.
3. **Pengfei Hu**, et al., “Discovering Hierarchy-Grounded Domains with Adaptive Granularity for Clinical Domain Generalization”, *Under Review*, 2025.
4. **Pengfei Hu***, et al., “AGFormer: Adaptive Spatiotemporal Graph Informed Transformer for Multi-Reservoir Inflow Forecasting”, *Environmental Modelling & Software (IF: 4.6)*, 2026.
5. **Pengfei Hu***, et al., “MPLite: Multi-Aspect Pretraining for Mining Clinical Health Records”, *IEEE International Conference on Big Data (IEEE BigData 2024)*.

Collaborative Publications:

1. Xiaoxue Han, **Pengfei Hu**, et al., “Interpretable and Interactable Predictive Healthcare with Knowledge-Enhanced Agentic Causal Discovery”, *Findings of the Association for Computational Linguistics (EMNLP 2025 Findings)*.
2. Hsin-Ling Hsu, ..., **Pengfei Hu**, et al., “MEDPLAN: A Two-Stage RAG-Based System for Personalized Medical Plan Generation”, *Annual Meeting of the Association for Computational Linguistics (ACL 2025, Industry Track)*.

HONORS

- Stevens Institute of Technology Provost Doctoral Fellowship
- University Outstanding Graduate (Summa Cum Laude) *Spring 2020*
- National Scholarship of Zhongli International (Top 0.5%) *Fall 2019*

PROFESSIONAL SERVICES

Teaching Experience:

- Teaching Assistant for CS-584: Natural Language Processing. *Fall 2024, Spring 2025*
- Teaching Assistant for ISE-535: Data Mining. *Spring 2022, Fall 2022*

Research Experience:

- Research Assistant in USC Viterbi School of Engineering. *Summer 2022*
- Research Assistant in Center for Economic and Social Research (USC Dornsife). *Fall 2022, Spring 2023*
- Graduate Researcher in Oak Ridge National Laboratory (GRO). *Summer 2025*
- Authorized User of the Frontier Supercomputer at the Oak Ridge Leadership Computing Facility.

Conference and Journal Review: NeurIPS 2026; ICML 2026; ACL Rolling Reviews (bimonthly since July 2025); KDD 2026; AAAI 2026; MICAI 2025; Computers in Biology and Medicine; IEEE Access.